

PRODUCT SENTIMENT ANALYSIS AND FEATURE ROADMAP PRIORITIZATION USING PYTHON

Case Study: FitLife Health & Fitness Applications

Submitted in partial fulfillment of the academic requirements
for the course/project assessment

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CERTIFICATE

This is to certify that the project titled “Product Sentiment Analysis and Feature Roadmap Prioritization Using Python” is an academic project carried out by **TUSHAR PILLAY** under my guidance and supervision. The work presented in this report is submitted as part of the requirements of the respective academic program.

The project demonstrates the application of Python-based data cleaning, natural language processing, sentiment analysis, TF-IDF feature extraction, and RICE prioritization to derive product and managerial insights from user reviews.

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Date: 12/09/26

Institute : SCMHRD

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide and the institute for providing the opportunity and guidance required to complete this project. I am thankful for the academic support, feedback, and resources that helped me understand and apply data analytics and natural language processing techniques.

I also acknowledge the publicly available Google Play Store app and user-review dataset used for this academic analysis. The project helped me strengthen practical skills in Python, data preprocessing, sentiment analysis, feature extraction, and data-driven product prioritization.

2. ABSTRACT

Mobile health and fitness applications receive large volumes of user feedback, making manual review analysis difficult. This project develops a Python-based analytical workflow to convert user reviews into actionable product insights and a prioritized feature roadmap.

The analysis uses Google Play Store app and review data, filters the Health & Fitness category, selects the top 30 applications by review count, and matches available user reviews. After cleaning, 272 reviews from 8 matched applications were included in the final analysis. VADER was used to classify review sentiment, while TF-IDF was applied to identify frequently important terms. The identified terms were then prioritized using a simplified RICE framework based on reach, impact, confidence, and effort.

The final analysis found 192 positive reviews (70.6%), 50 negative reviews (18.4%), and 30 neutral reviews (11.0%). The average review length was 147.33 characters. The highest-ranked RICE term was “app”, followed by “work”, “like”, “great”, and “day”. The results demonstrate how text analytics can support evidence-based product improvement and prioritization.

Keywords: Python, NLP, VADER, TF-IDF, RICE, Sentiment Analysis, Health & Fitness Apps, Product Analytics.

3. INTRODUCTION & PROBLEM STATEMENT

3.1 Background

Health and fitness applications depend heavily on user experience, engagement, tracking functionality, workout support, and ease of use. App-store reviews provide direct user feedback, but the volume and unstructured nature of this text make it difficult to identify common concerns and improvement opportunities manually.

This project addresses the problem by creating a structured Python workflow that cleans review data, analyzes sentiment, extracts important review terms, and converts those terms into a prioritized product roadmap.

3.2 Problem Statement

The central problem is to identify user sentiment and recurring feature-related themes from Health & Fitness application reviews and prioritize potential product improvements using a transparent analytical framework.

3.3 Objectives

- Clean and prepare Google Play Store app and review data for analysis.
- Classify user reviews into positive, negative, and neutral sentiment using VADER.
- Identify important terms and potential feature themes using TF-IDF.
- Prioritize identified terms using the RICE framework to support product decision-making.

3.4 Scope and Limitations

- Scope is limited to Health & Fitness applications represented in the selected Google Play Store dataset.
- Only 8 of the selected top 30 applications had exact app-name matches in the available review dataset, resulting in 272 final reviews.
- VADER may not fully understand sarcasm, context, mixed sentiment, or domain-specific language.
- TF-IDF terms are keywords and should not automatically be interpreted as complete product features.
- RICE Impact, Confidence, and Effort were held constant in this implementation; therefore, the ranking is primarily driven by Reach.

DATA SOURCE

Primary Data Source: Google Play Store Apps dataset by Lavanya (L. Gupta), hosted on Kaggle. The dataset contains googleplaystore.csv and googleplaystore_user_reviews.csv, used for application metadata and user-review analysis respectively.

Dataset URL: <https://www.kaggle.com/datasets/lava18/google-play-store-apps>

Recommended citation: L. Gupta, "Google Play Store Apps," Feb. 2019. [Online]. Available: <https://www.kaggle.com/lava18/google-play-store-apps>

4. LITERATURE REVIEW / EXISTING SYSTEMS

Sentiment analysis is a common natural language processing approach for converting unstructured user feedback into positive, negative, or neutral categories. Lexicon-based methods are useful when a lightweight and interpretable approach is required. VADER (Valence Aware Dictionary and sEntiment Reasoner) is designed for sentiment analysis of social-media-style and short textual content and provides a compound sentiment score.

TF-IDF is a standard information-retrieval technique that assigns higher importance to terms that are frequent in a document but relatively uncommon across the full collection. In product-review analysis, TF-IDF can be used to surface terms that may represent recurring user concerns, benefits, or product themes.

The RICE framework—Reach, Impact, Confidence, and Effort—provides a structured approach to prioritizing product initiatives. Combining text-derived terms with a prioritization framework can help move from descriptive review analysis toward product decision support.

4.1 Existing System

- Manual reading and categorization of app reviews.
- Basic app-store ratings without detailed text-level analysis.
- Standalone sentiment analysis without feature prioritization.
- Keyword analysis without a structured product prioritization framework.

4.2 Proposed Improvement

The proposed workflow integrates cleaning, sentiment classification, TF-IDF feature extraction, and RICE prioritization into one reproducible Python pipeline. This creates a clear path from raw reviews to product-oriented recommendations.

5. PROPOSED METHODOLOGY & SYSTEM DESIGN

5.1 Methodology

The project follows a sequential analytical workflow. App metadata and user reviews are loaded, cleaned, filtered to the Health & Fitness category, and reduced to the top 30 applications by review count. Matching reviews are then analyzed using VADER and TF-IDF. The most prominent terms are subsequently prioritized using RICE.

System Flow: Data Collection → Data Cleaning → Health & Fitness Filtering → Top 30 Selection → Review Matching → Review Analysis → VADER Sentiment → TF-IDF Feature Extraction → RICE Prioritization → Business Insights

5.2 Tools and Libraries

Tool / Library	Purpose
Python	Core programming and analysis
Pandas	Data loading, cleaning, filtering and tabulation
NumPy	Numerical operations
NLTK / VADER	Sentiment scoring and classification
Scikit-learn	TF-IDF feature extraction
Matplotlib / Seaborn	Output visualization
Jupyter / Google Colab	Development and execution environment

5.3 Input and Data Flow

Stage	Output
App dataset	Cleaned application metadata
Review dataset	Cleaned usable reviews
Category filter	Health & Fitness applications
Ranking	Top 30 applications by review count
Matching	272 reviews across 8 matched applications
VADER	Sentiment score and class
TF-IDF	Important review terms
RICE	Prioritized terms for roadmap discussion

5.4 Evaluation

Evaluation is based on dataset completeness checks, review counts, missing-value checks, successful VADER scoring, sentiment classification, TF-IDF output generation, and successful RICE score calculation.

6. IMPLEMENTATION

6.1 Dataset Loading

Two CSV files were used: the Google Play Store application dataset and the Google Play Store user-review dataset.

Key implementation:

```
apps = pd.read_csv('/content/googleplaystore.csv')
reviews = pd.read_csv('/content/googleplaystore_user_reviews.csv')
```

6.2 Data Cleaning and Filtering

Application records were cleaned by removing missing App, Category, and Reviews values, converting Reviews to numeric values, and removing duplicate app names. Review records with missing or very short translated reviews were removed, followed by duplicate removal.

```
health = apps[apps['Category'] == 'HEALTH_AND_FITNESS']
health = health[health['Reviews'] >= 100]
top30 = health.sort_values('Reviews', ascending=False).head(30)
final = reviews[reviews['App'].isin(top30['App'])].copy()
```

6.3 VADER Sentiment Analysis

VADER compound scores were converted into three classes: Positive for scores ≥ 0.05 , Negative for scores ≤ -0.05 , and Neutral otherwise.

```
final['VADER_Score'] = final['Translated_Review'].apply(
    lambda x: analyzer.polarity_scores(x)['compound']
)
```

6.4 TF-IDF Feature Extraction

TF-IDF was calculated using English stop-word removal and a maximum of 20 features. Mean TF-IDF scores were then used to rank the extracted terms.

```
tfidf = TfidfVectorizer(stop_words='english', max_features=20)
matrix = tfidf.fit_transform(final['Translated_Review'])
```

6.5 RICE Prioritization

The top 10 TF-IDF terms were assigned Reach based on the number of final reviews containing each term. Impact was set to 2, Confidence to 0.8, and Effort to 1. The score was calculated as $\text{Reach} \times \text{Impact} \times \text{Confidence} \div \text{Effort}$.

7. RESULTS & OUTPUT SCREENSHOTS

7.1 Dataset and Processing Results

Metric	Result
App dataset records loaded	10,841
App dataset columns	13
Review records loaded	64,295
Review dataset columns	5
Usable reviews after cleaning	27,307
Health & Fitness applications	254
Top applications selected	30
Applications matched to reviews	8
Final reviews analyzed	272
Average review length	147.33 characters

The data preparation stage reduced the review dataset to 27,307 usable reviews. After category filtering and top-30 selection, exact app-name matching produced 8 applications with available reviews. The final analytical dataset contained 272 reviews.

7.2 Review Length Distribution

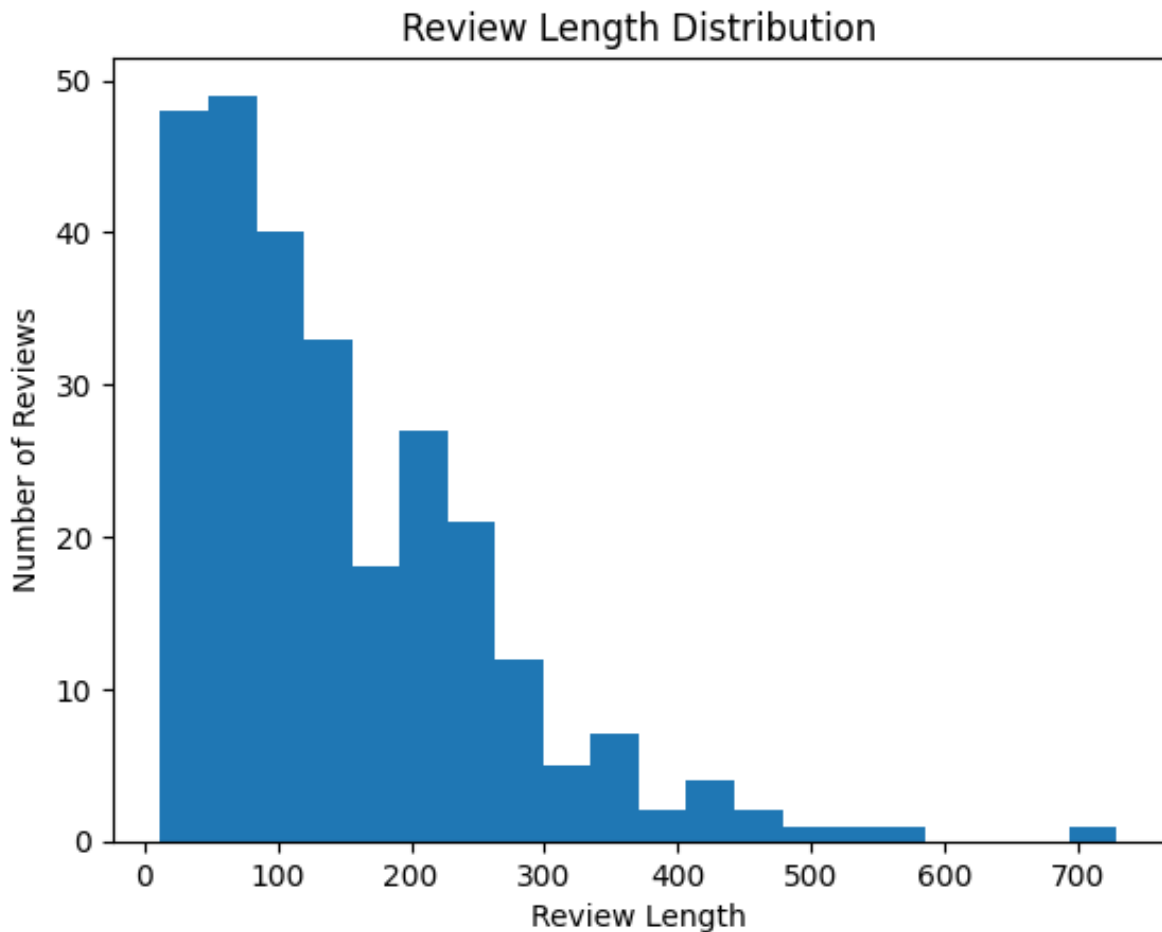


Figure 7.1: Review Length Distribution

The distribution is concentrated toward shorter reviews, with most observations below approximately 300 characters. A small number of longer reviews extend beyond 500 characters. The calculated average review length is 147.33 characters.

7.3 Sentiment Distribution

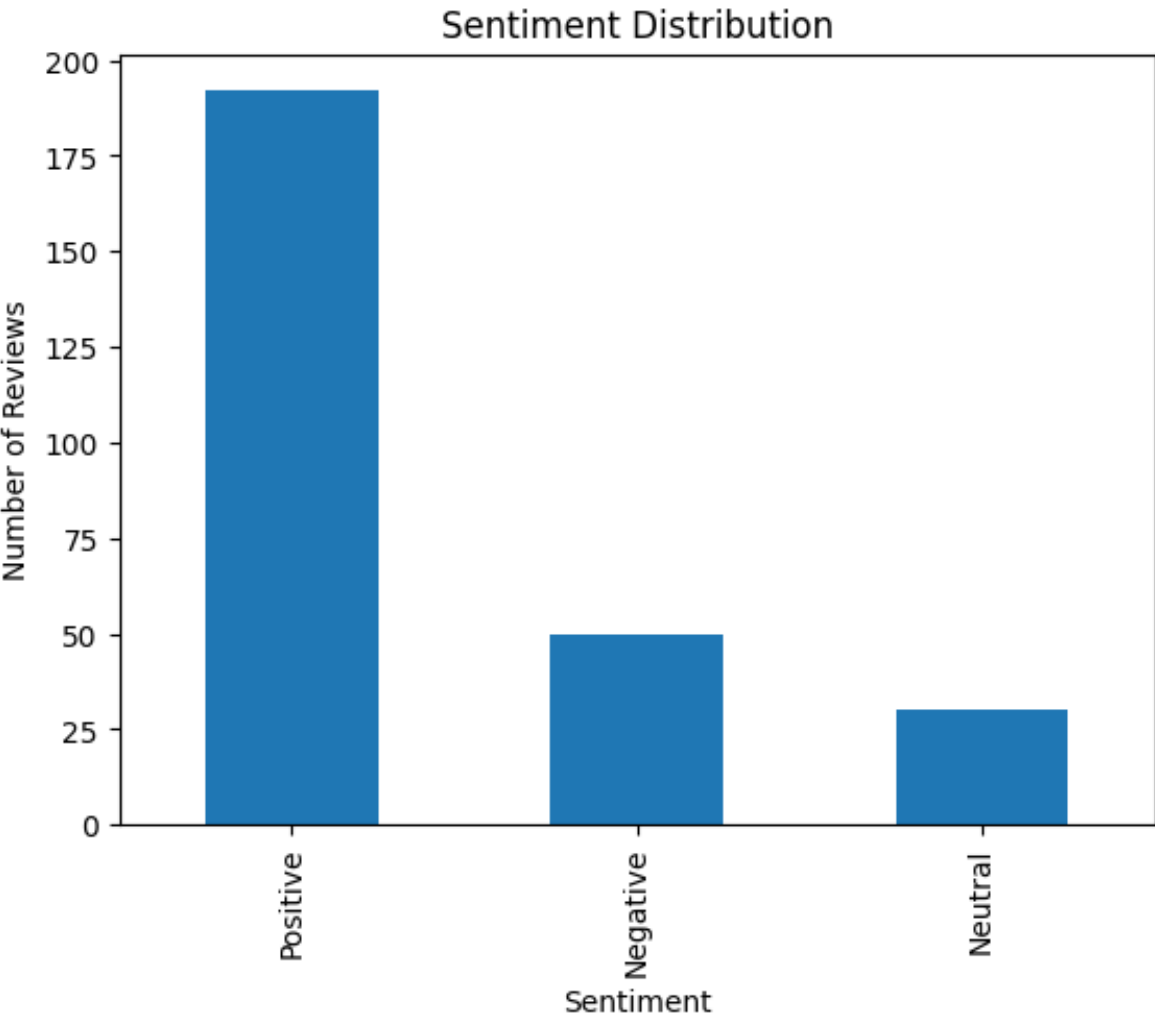


Figure 7.2: Sentiment Distribution

Sentiment	Reviews	Share
Positive	192	70.6%
Negative	50	18.4%
Neutral	30	11.0%

Positive sentiment dominates the final dataset, accounting for 70.6% of reviews. Negative sentiment represents 18.4%, while neutral reviews represent 11.0%. This indicates an overall favorable sentiment among the analyzed reviews, while the negative segment remains important for identifying improvement opportunities.

7.4 Sentiment by Application

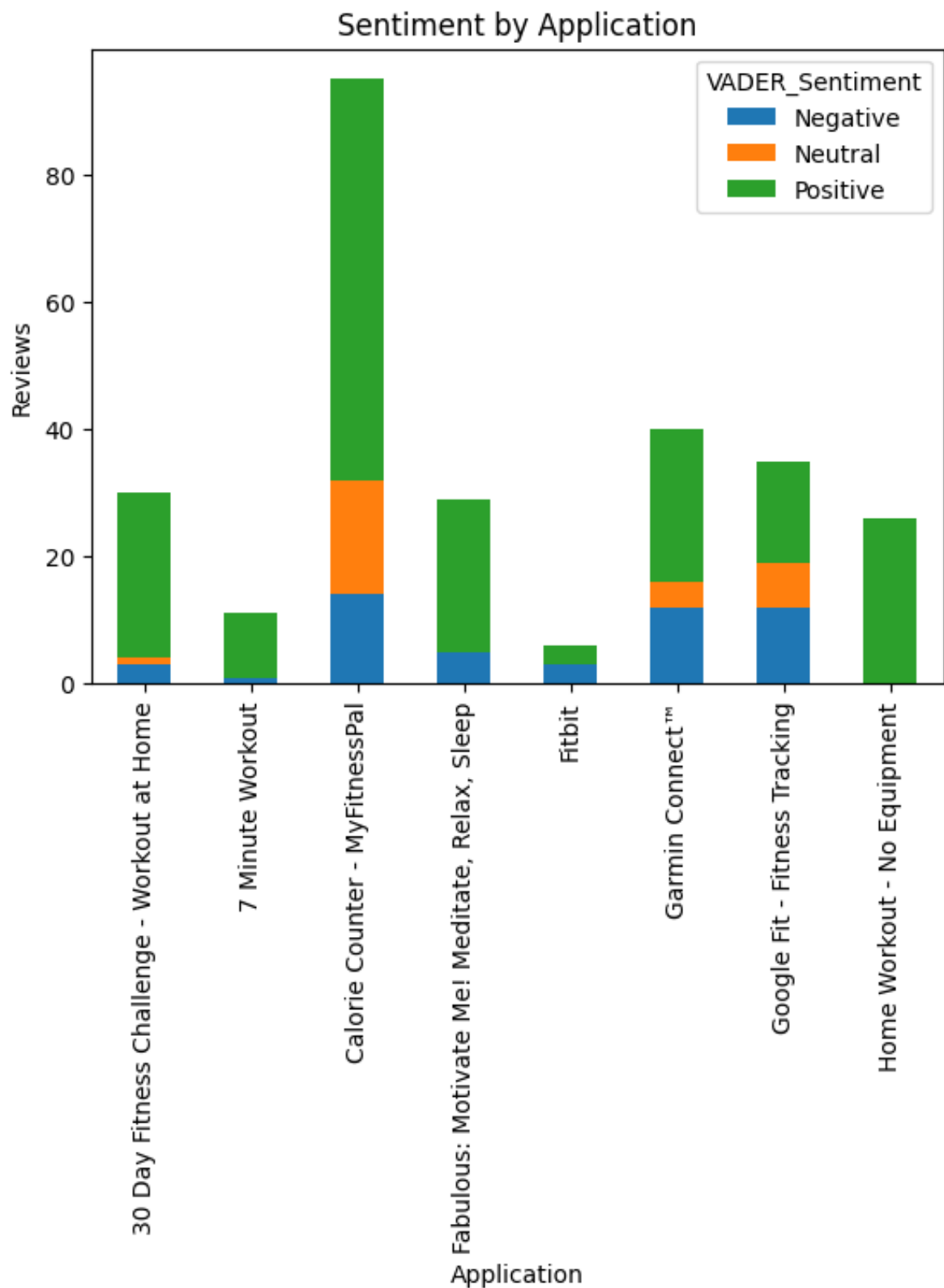


Figure 7.3: Sentiment by Application

The application-level output shows variation in the mix of positive, neutral, and negative reviews. Calorie Counter - MyFitnessPal has the largest review volume among the matched applications, while several applications show a predominantly positive review mix. This comparison helps identify applications with relatively stronger or weaker sentiment patterns.

7.5 TF-IDF Feature Analysis

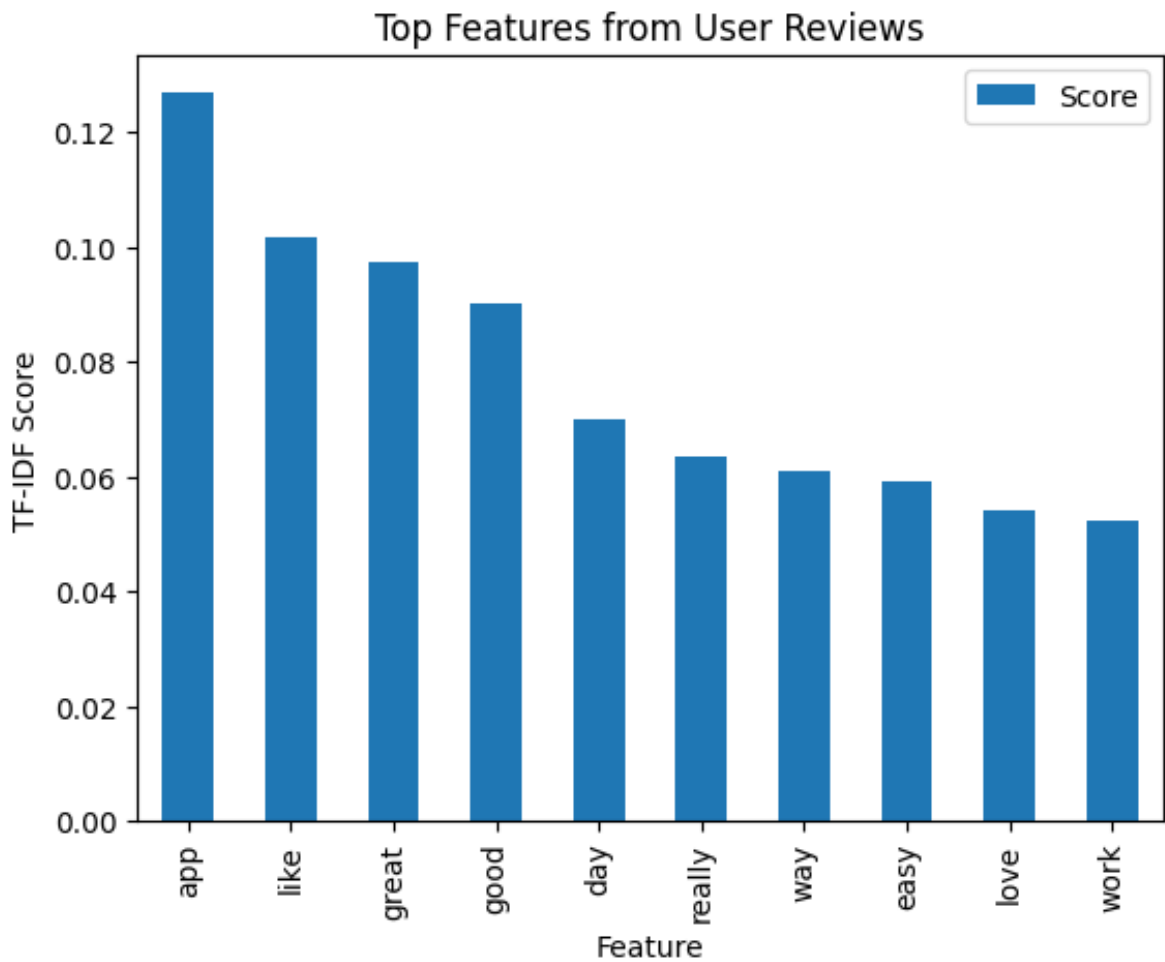


Figure 7.4: Top Features from User Reviews

The highest TF-IDF term is “app”, followed by “like”, “great”, “good”, “day”, “really”, “way”, “easy”, “love”, and “work”. These terms represent recurring language in the analyzed reviews. They should be interpreted as important textual themes rather than direct one-to-one product features.

7.6 RICE Feature Prioritization

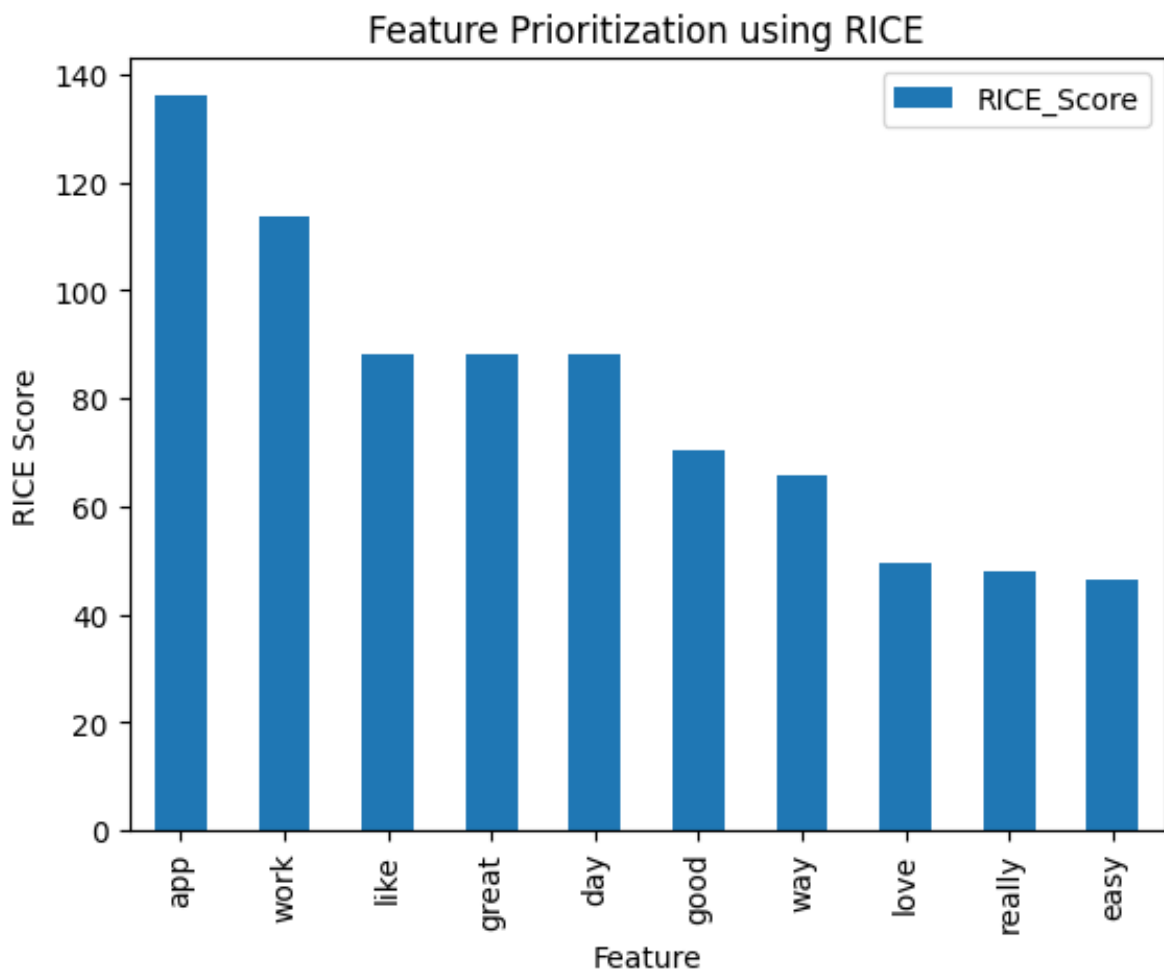


Figure 7.5: Feature Prioritization Using RICE

Rank	Feature / Term	Reach	Impact	Confidence	Effort	RICE Score
1	app	85	2	0.8	1	136.0
2	work	71	2	0.8	1	113.6
3	like	55	2	0.8	1	88.0
4	great	55	2	0.8	1	88.0
5	day	55	2	0.8	1	88.0
6	good	44	2	0.8	1	70.4
7	way	41	2	0.8	1	65.6
8	love	31	2	0.8	1	49.6
9	really	30	2	0.8	1	48.0
10	easy	29	2	0.8	1	46.4

The RICE ranking places “app” first because it has the highest Reach. Since Impact, Confidence, and Effort were kept constant, the resulting ranking is primarily a Reach-based prioritization. Therefore, the output should be treated as an initial prioritization aid rather than a final product roadmap.

7.7 Managerial Interpretation

- The strong positive sentiment indicates that many users perceive value in the analyzed applications.

- Negative reviews provide a focused opportunity for identifying usability and functionality pain points.
- Recurring terms such as “work”, “day”, and “easy” can guide deeper qualitative review of workout, routine, and usability experiences.
- RICE can help convert text-analysis findings into a structured prioritization discussion.
- Product teams should validate the keyword findings with qualitative review reading and domain knowledge before implementing roadmap decisions.

8. TESTING

Testing was performed at key stages of the analytical pipeline to ensure that the final dataset and generated analytical fields were complete and usable.

Test Case	Expected Outcome	Actual Outcome	Status
App dataset loading	CSV loads successfully	10,841 records loaded	Pass
Review dataset loading	CSV loads successfully	64,295 records loaded	Pass
Review cleaning	Remove missing/short/duplicate reviews	27,307 usable reviews	Pass
Health & Fitness filtering	Category filter works	254 applications	Pass
Top-30 selection	Rank by review count	30 selected	Pass
Review matching	Match selected app names	8 applications matched	Pass
Final dataset	Create analysis dataset	272 reviews	Pass
VADER scoring	No missing scores	0 missing scores	Pass
Sentiment labels	No missing classes	0 missing labels	Pass
TF-IDF extraction	Generate top features	20 features generated	Pass
RICE calculation	Generate prioritization scores	10 ranked terms	Pass

The testing results indicate that all major analytical stages completed successfully. The final dataset contained no missing VADER scores or sentiment labels.

9. CONCLUSION & FUTURE SCOPE

9.1 Conclusion

This project demonstrates a complete Python-based workflow for analyzing user reviews of Health & Fitness applications and translating unstructured feedback into structured product insights. The workflow combined data cleaning, VADER sentiment analysis, TF-IDF feature extraction, and RICE prioritization.

The final dataset contained 272 reviews from 8 matched applications. The sentiment analysis showed a predominantly positive pattern, with 192 positive, 50 negative, and 30 neutral reviews. TF-IDF identified recurring terms, while RICE provided a transparent ranking mechanism for initial prioritization.

Overall, the project shows how NLP and product prioritization techniques can support evidence-based decision-making. However, the RICE output should be interpreted cautiously because the current implementation uses fixed Impact, Confidence, and Effort values.

9.2 Future Scope

- Use more advanced semantic models such as transformer-based embeddings for improved contextual understanding.
- Perform aspect-based sentiment analysis to distinguish sentiment about specific features such as workouts, tracking, subscriptions, notifications, or usability.
- Replace constant RICE assumptions with product-manager or stakeholder-based estimates for Impact, Confidence, and Effort.
- Use broader or more recent review datasets to improve representativeness.
- Add automated dashboards for recurring review monitoring and product decision support.
- Use topic modeling or clustering to group similar review themes before roadmap prioritization.

10. REFERENCES & APPENDIX

10.1 References

- Gupta, L. (2019). "Google Play Store Apps." Kaggle. <https://www.kaggle.com/datasets/lava18/google-play-store-apps>
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10.2 Appendix A – Full Source Code

Python Source Code File: [Open Python / Project Files Google Drive Folder](#)

Excel / Downloaded Data Files: [Open Excel and downloaded data files in Google Drive](#)

Note: The Google Drive folder contains the project source files and downloaded data files. The Excel/data files should be retained with the project submission as supporting data.

The complete Python source code used for data loading, cleaning, filtering, sentiment analysis, TF-IDF extraction, RICE prioritization, testing, and CSV export should be attached/pasted here as required by the faculty.

Core workflow used in the project:

- Import libraries
- Load Google Play Store app and review datasets
- Clean application metadata
- Clean review text
- Filter Health & Fitness applications
- Select top 30 applications by review count
- Match available reviews
- Calculate review length
- Run VADER sentiment analysis
- Generate sentiment-by-application output
- Extract TF-IDF features
- Calculate RICE scores
- Run validation/testing checks
- Export final analysis CSV files

Project Data & Source Files Folder:

https://drive.google.com/drive/folders/1CtfboqwPOq2IKGxUG93sk82Ex1kxanrP?usp=drive_link